

Understanding Lucidity - Science & Practice

Jonathan Berent
CEO, NextSense, Inc



About Me



The Moon shoot Factory
Project Lead



Lucid Reality
CEO, Lucid Reality



Stanford University

Guest Lecturer
Sleep & Dreams



Researcher
2 Way Communication
during REM sleep
(Cell Biology)

Jonathan Berent

CEO, NextSense

Win a free pair of NextSense Smartbuds by asking questions!





Questions

- How do we know someone is actually having a lucid dream?
- Is dream time the same waking time?
- Is a lucid dream an altered state of consciousness or just a phase of sleep?
- Is dreaming more similar to or different from imagination?
- Can you decode through AI what someone is dreaming without them describing it?
- What are some of the cutting edge areas of research for LD?

Lucid Dreaming definition:

A dream where the individual is **aware** they are dreaming. The dreamer **may** be able to exert some level of control over the dream narrative, dream content, and environment.





Dreams lead to reality



Dream Composing



"For about a month I went round to people in the music business and asked them whether they had ever heard it before. Eventually it became like handing something in to the police. I thought if no one claimed it after a few weeks then I could have it." - Paul McCarthy (source Cross, Craig (2005). *The Beatles: Day-by-Day, Song-by-Song, Record-by-Record*. Lincoln, NE: iUniverse, Inc. ISBN 0-595-34663-4)



Dream State
Dana Leong

This is an entire album that was composed by noting themes that I journaled after sleep / dreams. Two-time Grammy winning musician, Dana Leong.
[@danaleonmusic](#)



Dreams

lead to reality

..when I suddenly woke up, I was thinking: what if we could download the whole web? I grabbed a pen and started writing....when a really great dream shows up, grab it!"

-Larry Page

FIRST DAY COVER

BY AIR MAIL
PAR AVION



S. Ramanujan
1887-1920

Prof. DAVID BORWEIN,
Dept. of Mathematics,
University of St. Andrews,
10 West Acres,



Infinite Series of π

Mathematical Significance



Rapid Convergence: Used practically in numerical computations of π .



Mystery and Elegance: The series is linked to deep areas in mathematics such as modular forms and elliptic curves, discovered decades later.



G.H. Hardy is often credited with 'discovering' Ramanujan after receiving a letter full of extraordinary theorems—many without proofs. Hardy invited him to Cambridge and became a mentor of sorts, helping Ramanujan translate his intuitive insights into rigorous mathematical form.

$$\frac{1}{\pi} = \frac{2\sqrt{2}}{9801} \sum_{k=0}^{\infty} \frac{(4k)(1103 + 26390k)}{(k!)^4 396^{4k}}$$



Intuitive, Dreamlike Discovery

Ramanujan reported these insights as divine, appearing almost fully formed rather than derived through deliberate calculation. He often claimed his results were revealed by the goddess in his dreams, underscoring the mysterious, intuitive origins of his profound mathematical discoveries.



What makes lucid dreaming special:



Self-Discovery:
Access unconscious
parts of your mind.



Enhanced Creativity:
Unlock new ideas
and inspiration.



Personal Fulfillment:
Increase your sense
of well-being.

All without psychedelics or extensive psychotherapy.

**What if you could dream about any
topic or experience on demand?**

**What if you could exert some measure
of control or volition to your dreams?**



1. LUCID DREAMS OCCUR DURING THE REM SLEEP CYCLE



Eye movements are voluntary in the dream state and can be observed externally.

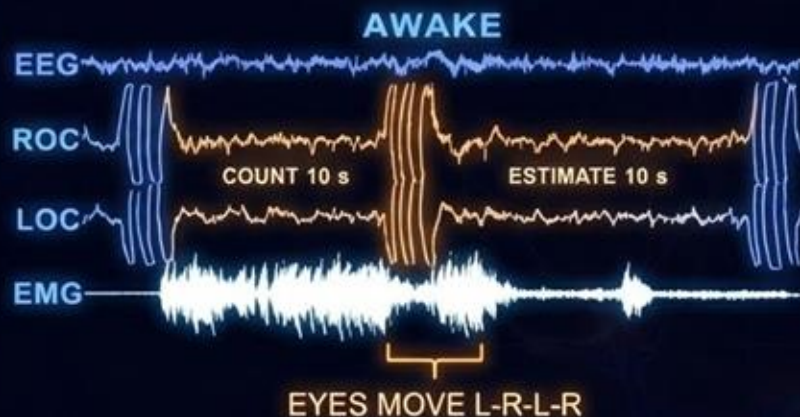


Dream tasks include counting to 10 or signaling Left-Right-Left-Right inside a dream.



Many physiological signals correspond accurately to dream content (e.g. heart rate, sexual response).

LUCID DREAMING RESEARCH @ STANFORD

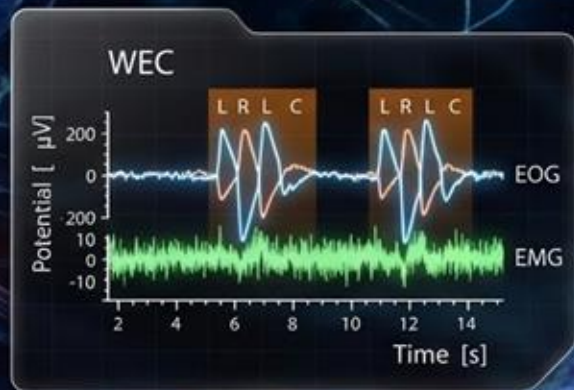


LaBerge et al.

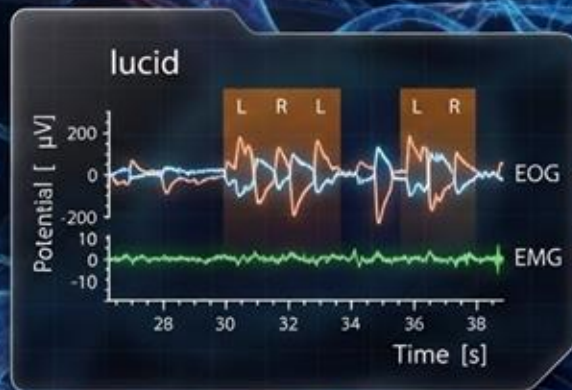
ADOPTION OF LABERGE'S LRLR EYE SIGNALING METHODOLOGY

Lucid Dreaming Research @ Stanford

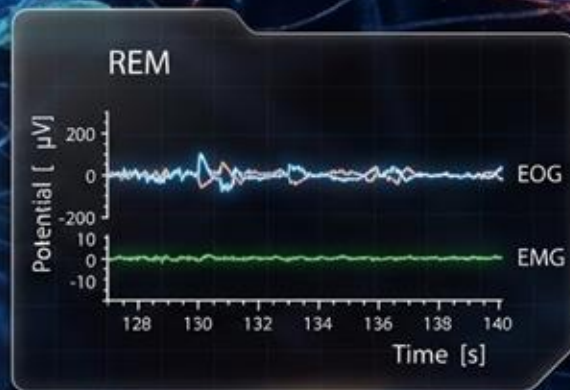
Voss, U., Holzmann, R., Tuin, I., & Hobson, A. J. (2009). *Sleep*



WEC



lucid



REM

LUCID DREAMING PROPOSED AS “HYBRID” OF REM AND WAKE STATES

“In order to move from non-lucid REM sleep dreaming to lucid REM sleep dreaming, there must be a shift in brain activity in the direction of waking.”
(Voss, U., Holzmann, R., Tuin, I., & Hobson, A. J. (2009). *Sleep*)

WAKING EYES CLOSED (WEC)



Increased high frequency (HF) EEG power in frontal lobe (40Hz)



Heightened cortical networking across short, middle, and long ranges for all frequency bands



Similar to LD EOG with more pronounced spikes



High EMG activity

LUCID DREAMING (LD)



Large amplitude, repetitive EOG spikes during Lucid eye-signalling



Heightened cortical networking across short, middle, and long ranges for all frequency bands



Large amplitude, repetitive EOG spikes during Lucid eye-signalling



Mean EMG amplitude showed no systematic variability between the 2 states

REM SLEEP



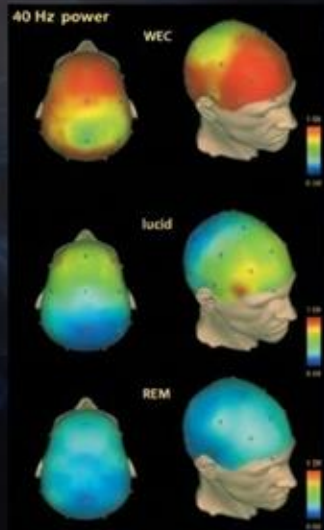
Diminished HF EEG power globally



Minimal cortical networking

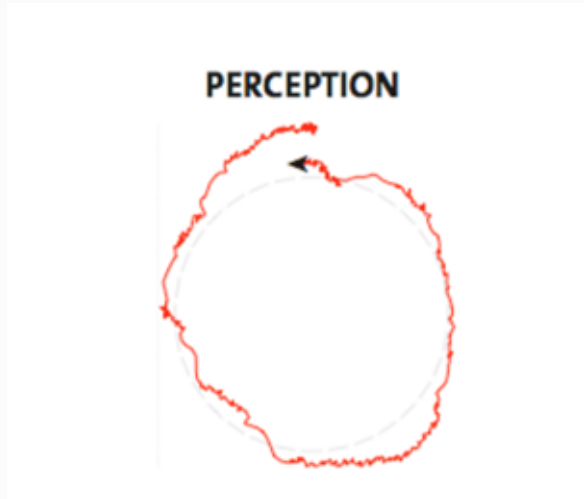


Lower amplitude and variable EOG spiking

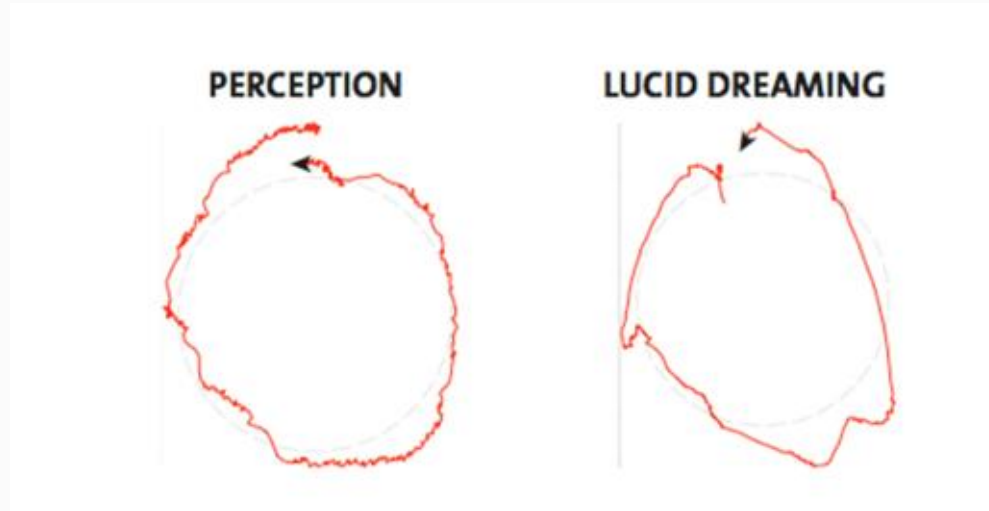


“[...] observed increases in frontal lobe 40-Hz power and the global increase in EEG coherence compared to REM sleep may be signs of a distinctive neurophysiology.”

#2: Lucid dreaming is more tightly correlated to perceptive faculties than to imagination



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#2: Lucid dreaming is more tightly correlated to perceptive faculties than to imagination

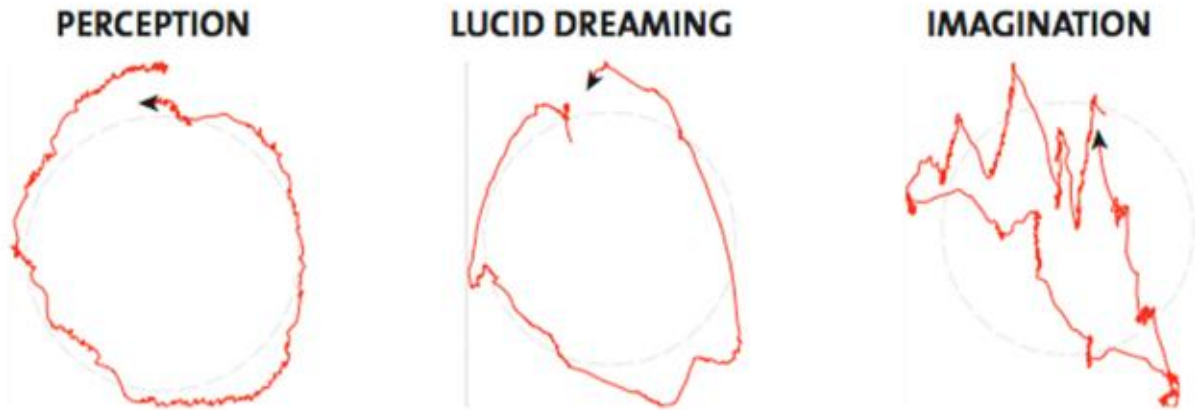
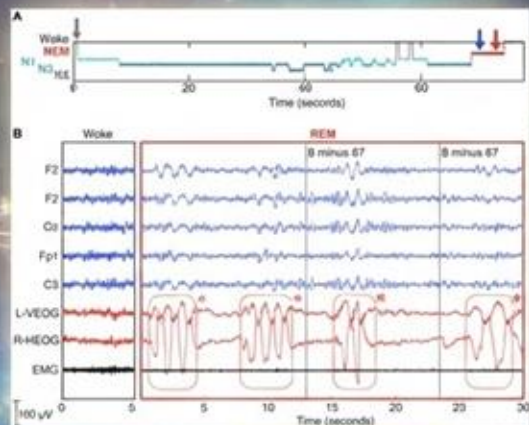


Figure 5. Eye-movement tracking of a circular movement compared in perception, lucid dreaming, and imagination. Note the relatively smooth tracking of perception and dreaming contrasted with the jerky saccadic eye movements of imagination.

Two-way communication is possible within a **dream state**

Konkoly, Berent, Paller, et al, 2021 Cell Biology



- Figure A: REM started 68 min into the recordings (auditory cue at blue arrow)
- Figure B: LRLR signaling lucidity (red *) were followed by spoken auditory stimulus "8 - 6"
- The correct answer was given with 2 eye signals.
- This was repeated again successfully.

Executive Summary of Study Design



Experiment Participation:

American, German,
French, Netherlands



Participants:
36 participants

Sessions (82 total):



57 had REM



15 had signal
verified lucid
dreaming.
(¾ raters had to
agree on whether
a signal was
present).

Trials (158 total):



29 correct answer



5 incorrect



28 ambiguous



96 non-responses

Example of a participant solving the math problem - "8 - 6" with eye movements

Current Biology
Article

CellPress
OPEN ACCESS

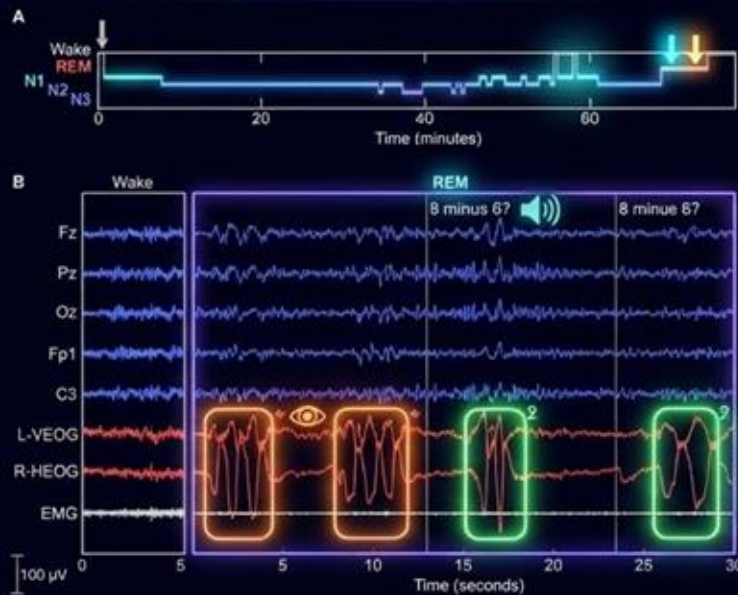


Figure A: Lucid Dreaming Communication (EEG/EOG)

REM started 68 min into the recordings (auditory cue at blue arrow)

Figure B:

LRLR signaling lucidity (red *) were followed by spoken auditory stimulus "8 - 6".

The correct answer was given with 2 eye signals.

This was repeated again successfully.

#4: TWO-WAY COMMUNICATION IS POSSIBLE WITHIN A DREAM STATE (CONT)

Konkoly, Berent, Paller, et al.



**USA
TEAM**



Task: Math Problems

Total Trials: 31

6

Correct
Responses

1

Incorrect

5

Ambiguous

19

No Responses

GERMANY TEAM

Tasks: Math, Morse Code
& Eye Movements

Total Trials: 54

1

Correct Response

FRANCE TEAM

Tasks: Counting,
Discrimination & Yes/No

Total Trials: 65

21

Correct Responses

NETHERLANDS TEAM

Task: Math Problems

Total Trials: 8

1

Correct Response

TOTAL: 29 Trials

Correct (18.4%)

5

Incorrect (3.2%)

28

Ambiguous (17.7%)

96

No Responses (60.8%)

Exciting New Areas of Research

BCI from within a LD



OTHER KEY ACID-BASE PAIRS

bioRxiv preprint doi: <https://doi.org/10.1101/2017.08.29.182294>; this version posted September 1, 2017. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

BSI from within a sknattal intense emmoutator to interase
woostain in vittuin comnuhense asoatocs

Support: R. Arnold, J. Dumas & University¹, Thomas J. McInerney², Roger A. Kinnel³, Katherine H. Dunham⁴,
Thomas R. Linder⁵, and Grant R. Gerson⁶

Hypnotic suggestion for LD induction



Placebo Experimentation inside an LD



ORIGINAL RESEARCH PAPER

9240

PLACEBO WITHOUT DECEPTION

[illegible]

Exciting New Areas of Research, cont.

Mindfulness & LD



Investigating the relationship and potential benefits of mindfulness practices on inducing and enhancing lucid dreaming.

Sleep Paralysis & LD state comparison



Exploring the phenomenological and neurophysiological overlaps between sleep paralysis and lucid dreaming states.

Brain Disorder Research with LD



Studying lucid dreaming's potential for understanding and treating conditions like psychosis and other brain disorders.

Replicated Research - Good Trend to Watch

Enhanced Sports Performance



Practicing skills in lucid dreams can improve real-world performance.

PTSD & LD Therapies



Lucid dreaming as a therapeutic tool for trauma and nightmares.

System Review of Induction Techniques



Evaluating effectiveness of methods to induce lucidity.

Top Researcher in the Field - Martin Dresler



h-Index



Funding



Number of grad students (size of lab)



Conferences / Intangibles

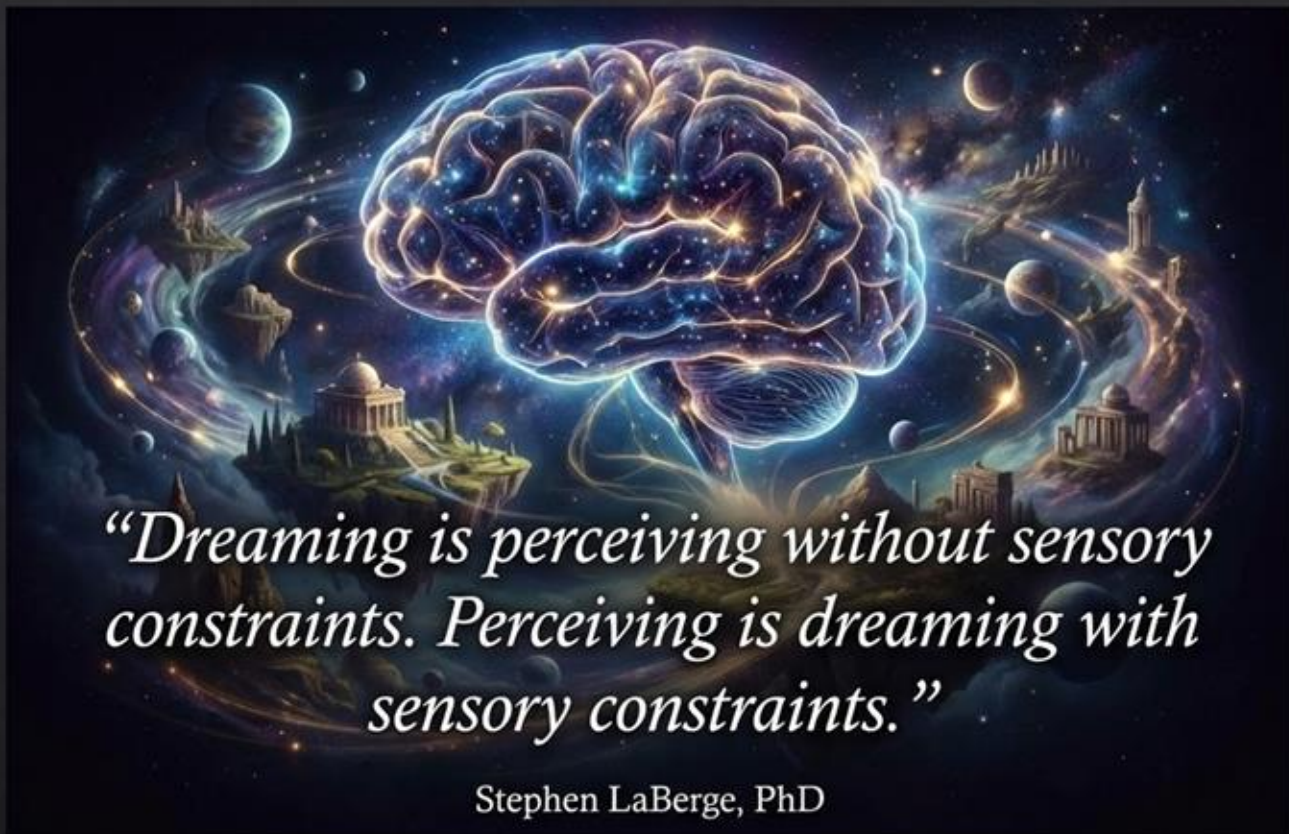


Novelty of research / contribution to the field



Figure 1: Lucid dreaming is a rare state of mind, however its frequency can be increased through multiple induction strategies. Induced eye signals during sleep inhibit the occurrence of lucid dreaming episodes (see top right panel, surrounding electrophysiological recordings of lucid dream onset with electroencephalogram). This objective signal is used to identify lucid dream onset (and other dream content) in time helps to uncover the neural mechanisms underlying the phenomenon, which in turn can inform induction strategies. Despite being discussed in the literature for centuries, and neuroscientifically studied for decades, techniques such as the reality and instability of lucid dreams have led to a relatively slow growth of the field. However, increasing progress in overcoming these constraints has produced considerable momentum in recent years. The ongoing standardization of research procedures, large-scale open science collaborations, emerging neurotechnology, online neuroscience, and in dream dream interfaces have the potential to further accelerate this development.

How are lucid dreams so lifelike?



“Dreaming is perceiving without sensory constraints. Perceiving is dreaming with sensory constraints.”

Stephen LaBerge, PhD

A couple is shown lying in bed, appearing to be asleep. The scene is framed within a dark, arched opening, suggesting a window or a portal. Outside this frame, a large, bright blue and white planet, resembling Earth or a moon, dominates the night sky. The background is a deep space filled with stars and nebulae. The entire image is overlaid with a complex network of glowing blue and orange lines that resemble neural connections or a dream map, with several bright points of light at the intersections.

DREAM SEX IS HIGHLY REALISTIC & CAN RESULT IN ACTUAL PHYSICAL ORGASMS

Research by psychophysicologists Stephen LaBerge and Walter Greenleaf (1983) recorded the first lucid dream orgasm. Subject Beverly D'Urso used pre-agreed eye movements to signal orgasm onset in her lucid dream. Simultaneously, her vaginal blood flow, muscle activity, and respiration rate reached their highest levels. - *Psychology Today* (2018).



LaBerge, S., W. Greenleaf, and B. Kedzierski. "Physiological Responses to Dreamed Sexual Activity during Lucid REM Sleep." *Psychophysiology* 20 (1983): 454-455

Key Takeaways: The Science of Lucid Dreaming

01 Lucid Dreams typically happen in the REM sleep cycle.

02 Lucid dreams can be used for self-discovery, enhancing creativity, and personal fulfillment.

03 The scientific consensus for signaling a lucid dream is L R L R eye-movements during the dream.

04 Lucid dreaming is a hybrid of wake and REM states

05 Two-way communication is possible within the dream state

Insights from advanced neurobiology and sleep hacking research.

Are we ready for some practical
advice on how to lucid dream?

Anyone want some advanced techniques?



Are there any downsides to lucid dreaming?



Fragmented sleep
is a marker of poor
sleep quality



Dissociative elements of lucid dreaming
are being studied in the context of psychosis



Insight, dissociation, control are defining
characteristics of lucid dreaming (Voss, Hobson,
2018 *Frontiers in Psychology*)



EEG markers (gamma band attenuation) for
dissociation and for insight (gamma band activation)



Difficult to study
clinically or
scientifically...

**It takes a lot of sleep to be a regular lucid dreamer -
overall benefits may outweigh the risks.**

Ethical & Philosophical Implications



- Psychological Risks: Dissociation, Fragmented Sleep, and Psychosis.



- Therapeutic Potential: Used for PTSD and Nightmare Reduction.



- Escapism: Lucid Dreaming could be used to avoid reality.



- Consciousness: Lucid Dreaming provides insights on the nature of consciousness.



My Experience with Lucid Dreaming



-  Started my first lucid dream journal when I was 18
-  First book - Lucid Dreams in 30 Days, Harary & Weintraub
-  Freshman Thesis on Lucid Dreaming (Dr. Stephen LaBerge)
-  Hosted Dr. LaBerge at Google
-  Attended Dream Yoga workshop w/Rinpoche Tenzin Wangyal
-  76 Lucid dreams recorded over 14 months (2011 - 2012)
-  My last lucid dream Jan 13, 2026

Source – JB Berent, “JB dream journals”

Strong correlation between lucidity level and significance



- **Strong positive correlation** (0.67) between lucidity level and dream significance.
- Highly lucid dreams tend to be **more personally meaningful**.

Source - JB Berent, "JB dream journals"

WILD most effective dream induction method



Induction Methods Effectiveness:



WILD (Wake-Induced Lucid Dreaming):
26 instances



DILD (Dream-Induced Lucid Dreaming):
21 instances



MILD (Mnemonic-Induced Lucid Dreaming):
3 instances

Source - JB Berent, "JB dream journals"

Lucid Dreaming Levels



Level 1 - Recognize you are lucid and do not wake up immediately (*beginner*)



Level 2 - Explore all five senses (*beginner*)



Level 3 - Remember a dream task you want to perform (*intermediate*)



Level 4 - Summon an object or a person to meet (*intermediate*)



Level 5 - Travel to a location you desire (*advanced*)



Level 6 - Meditate inside of the dream (*advanced*)



Level 7 - Dream Prompting (any level)

• **"Show me what my concept of God looks like"**

• **"Show me something beautiful"**

• **"Show me an advanced technology"**



Research supports learnability

Lucid Dreaming as a Learnable Skill



- Learnable skill
- Dream recall is critical
- Some techniques perform better than others
- Motivation is the key
- Applications range from nightmare reduction to rehearsals

Dream Journal - Week 1

Building the Foundation of Recall



- Buy a notebook just for recording dreams
- Tell yourself before going to bed "I want to remember my dreams."
- Set the intention to not move, or open your eyes, when you first wake up
- When you first awake, ask yourself, "What was I just dreaming?"
- Journal either by voice or write / type the dream. Notice any "dream signs" (things that are repeatedly showing up)

Recommended you get to 2 to 3 dreams per night before advancing to the next stage.

Reality Checks (RC) - Week 2

The more personalized the better.....



- Hold your nose and try to breath
- Look at your hands
- Read text, look away, read it again
- Imagine yourself floating
- Ask yourself, "Am I dreaming?"

Research suggests this technique is not effective by itself and must be practiced 3 - 8 weeks to see results. (Purcell 1988, 1986, Schlages 1992)

Dream Induced LD (DILD) - Week 3

Dream Induced Lucid Dreaming



- Review your “dream signs” from your dream journal
- Tell yourself before you go to sleep to recognize dream signs in your dreams. “If I see X, I will know I am dreaming...”
- If you see your dream signs during the day, perform a “reality check”
- Question reality during the day, especially when something bizarre happens, imagine that it IS a dream
- When you wake up in the morning, rehearse your dream and imagine that you did get lucid, be specific how you did

MILD - Week 3.5

Mnemonic Induction of Lucid Dreams



- Pioneered by Dr. Stephen LaBerge @ Stanford
- Mantra: "Next time I'm dreaming, I will remember I'm dreaming"
- Visualize the dream you want to have
- Visualize common dream signs
- Don't open your eyes when first wake up

* Only 2 studies independent of its creator, yet they suggest effectiveness of the method. (Aspy 2017)

WBTD / WILD Week 4

Wake-back-to-bed / Wake Induced Lucid Dreaming



- WBTD + RC + MILD most effective (Aspy, 2017)*
 - ✧ 1. Set an alarm for 5 hours
 - ✧ 2. See if you were dreaming (replay the dream)
 - ✧ 3. Stay up for 5 - 10 min, read about lucid dreaming, script a dream, or anything connected with lucid dreaming (read Comatose)
 - ✧ 4. As you fall asleep repeat to yourself, "The next time I am dreaming I will remember that I am dreaming."

Most effective if you fall asleep within 5 min (Aspy, 2017)

* WBTD + RC + MILD resulted in 53% experiencing at least 1 LD during week 2 of a comprehensive study (Aspy 2017)

SSILD Week 4.5

Senses Induced Lucid Dreaming



- Cycle through your 5 senses
 - Touch
 - Taste
 - Sight
 - Sound
 - Smell
- Once, slowly (20 - 30 sec per sense)
- Then **4 to 5 times** before sleep onset

Anyone want to share a dream?



Some tips for recording dreams:

- ☀ **Title your dream (try it now, for those who had a dream put a title in the chat box)**
- ☀ **Describe your feeling tone (happy, excited, anxious, frustrated, low / high)**
- ☀ **Pay special attention to characters you know / have met**
 - **What do you associate with this person?**
 - **Strengths? Weaknesses?**

Remember: There is no "wrong" or "right" interpretation.

DREAMTIME ROUTINE



No electronics 1 hour before bed.



Avoid caffeine after 2pm.



Designate a journal for your thoughts and dreams, and a special pen.



Repeat, eyes closed, "Tonight, I want to remember I'm dreaming when I'm dreaming."



60 seconds, now before opening eyes, rehearse what was going through your mind

REALITY TESTING AND THE MNEMONIC INDUCTION

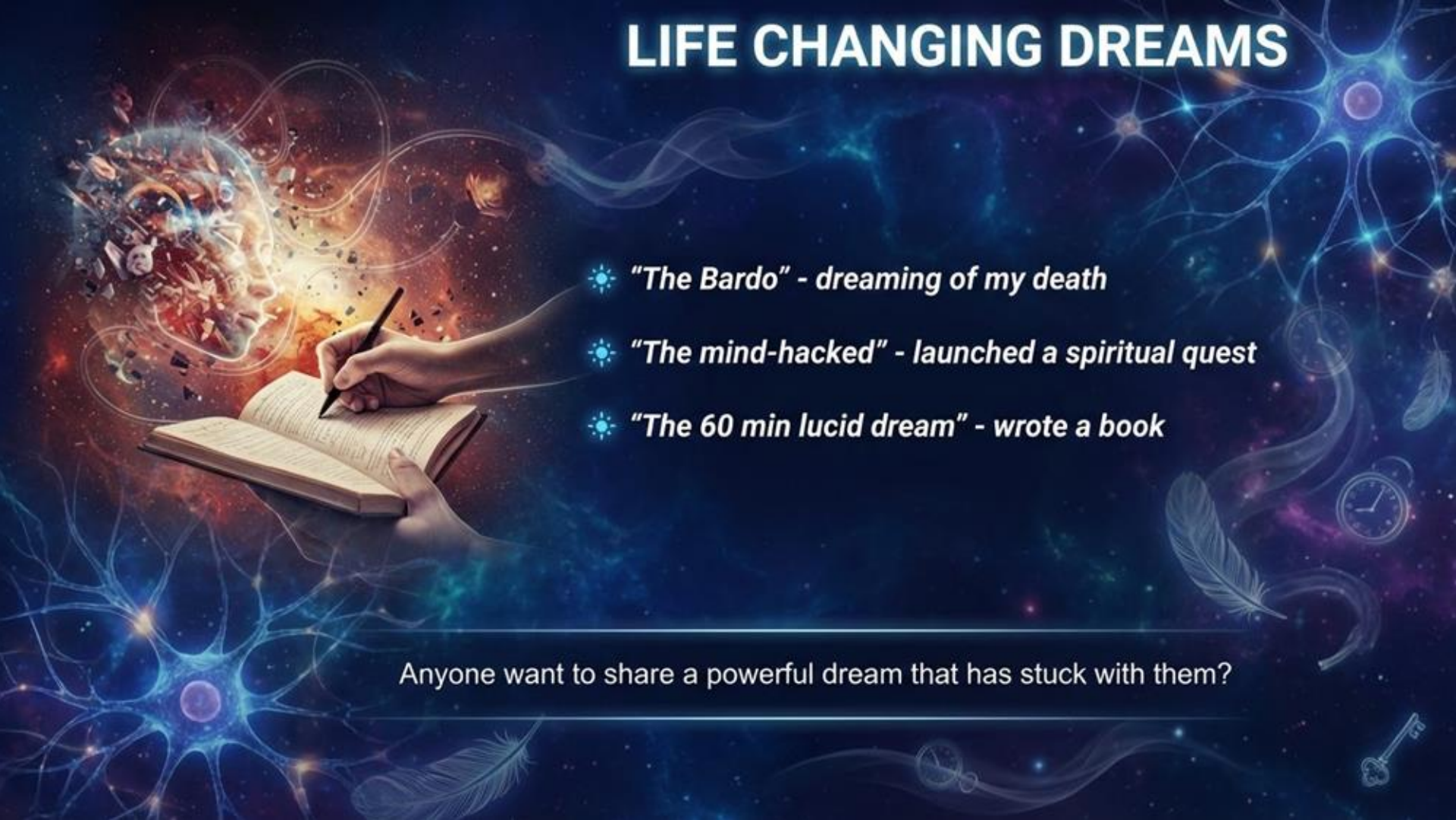
Table 4

Improvements in Lucid Dreaming Rates in Week 2 Compared With Week 1 for All Participants Combined and for Participants in Each Week 2 Group

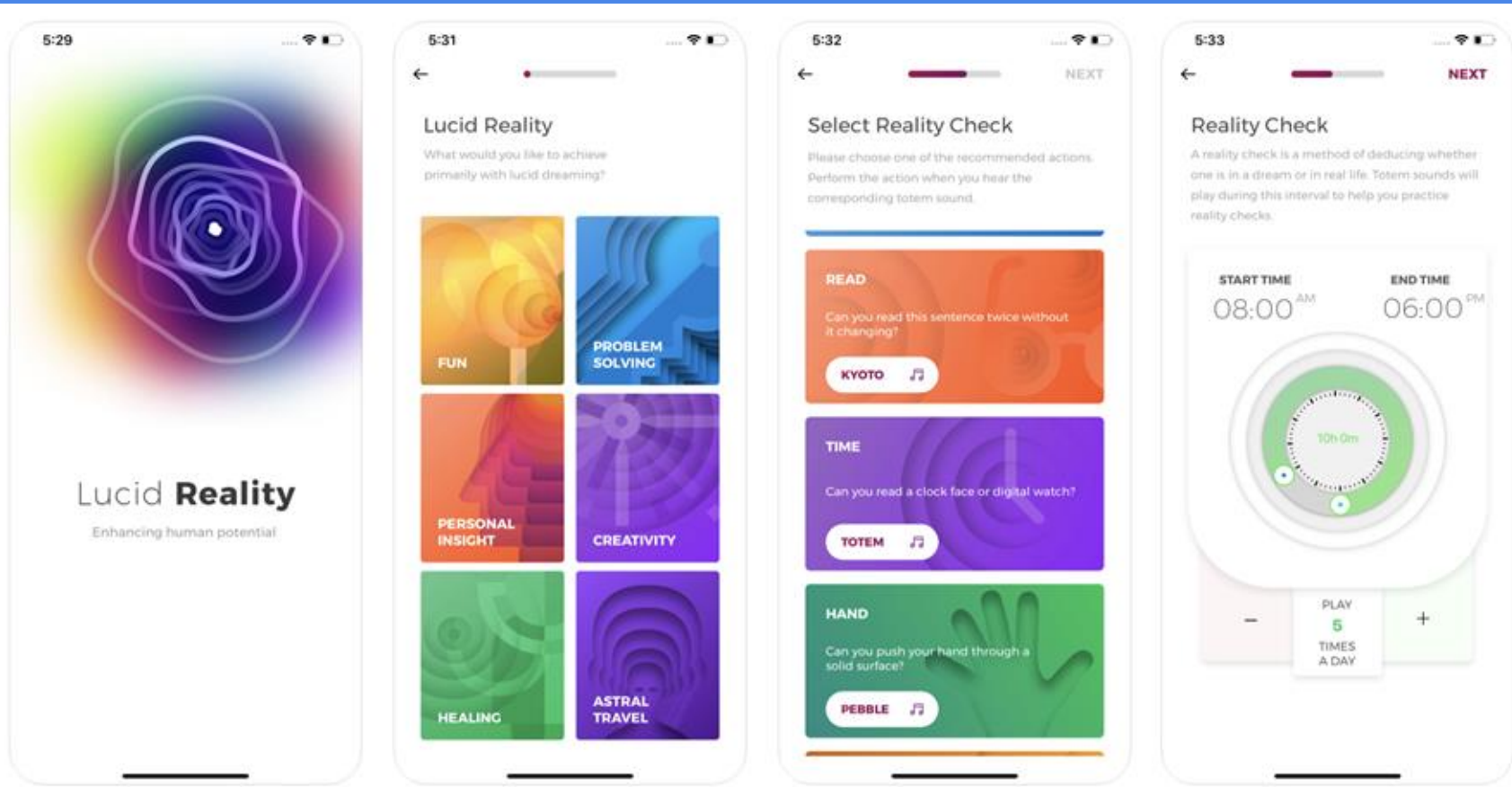
Week 2 group	L DRF Lucid			Wilcoxon test	
	Week 1 <i>M (SD)</i>	Week 2 <i>M (SD)</i>	Improvement	<i>Z</i>	<i>p</i>
All participants (<i>n</i> = 169)	8.1% (17.8%)	11.3% (17.3%)	39.0%	2.27	.023
RT only (<i>n</i> = 68)	8.1% (17.8%)	7.6% (13.0%)	-6.8%	.27	.786
RT + WBTB (<i>n</i> = 54)	6.9% (15.7%)	10.7% (16.3%)	54.1%	1.04	.301
RT + WBTB + MILD (<i>n</i> = 47)	9.4% (20.0%)	17.4% (22.0%)	84.5%	2.94	.003

Note. L = logbook variable; RT = reality testing; WBTB = wake back to bed; MILD = mnemonic induction of lucid dreams; DRF = dream recall frequency.

LIFE CHANGING DREAMS

- 
- The background of the slide is a dark, cosmic blue space filled with glowing blue neural network patterns, resembling brain cells. On the left, a hand is shown writing in an open book. Above the book, a glowing, translucent head is depicted, composed of various symbols and patterns, suggesting a dream or a complex thought process. The overall aesthetic is ethereal and dreamlike.
- ☀ *"The Bardo" - dreaming of my death*
 - ☀ *"The mind-hacked" - launched a spiritual quest*
 - ☀ *"The 60 min lucid dream" - wrote a book*

Anyone want to share a powerful dream that has stuck with them?

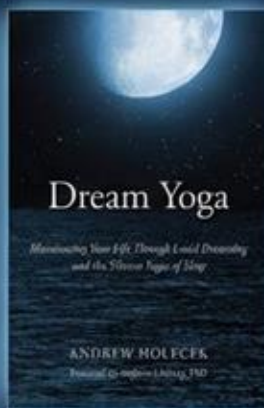
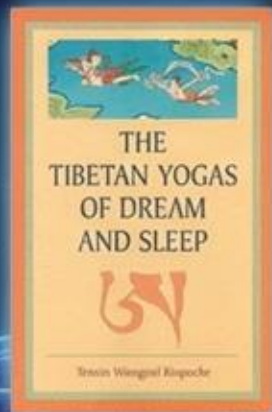


Intention + reminders are key

Lucid Reality, iOS App Store (Berent, Paller)

Lucid Dreaming Resources

Recommended Reading



Digital Resources & Communities



Get an App!



Lucidity.com (oldest/reliable website)
Dreamviews.com (active community)



Blogs: World of Lucid Dreaming,
themindhacked.com

Thank you!

jb@nextsense.io

Connect with me on LinkedIn

Or shoot me an email.